

On Fine-Tuning

7. What does the literature say about catastrophic forgetting when fine-tuning large pre-trained ViTs on small datasets?
8. Why does LP-FT outperform naive end-to-end fine-tuning, and under what conditions?
9. What alternatives exist (LoRA, BitFit, adapters) and why did we not use them?

Yes, catastrophic forgetting is real in small-data ViT fine-tuning, and LP-FT usually beats naive FT when pretrained features are already strong.

The literature here splits into **forgetting in small-data ViT fine-tuning**, **why LP-FT helps**, and **what PEFT alternatives exist and why they were not the best default choice**.

Catastrophic Forgetting

Large pretrained ViTs appear unusually vulnerable to **feature drift** and overfitting when fully fine-tuned on small downstream datasets (Liu et al., 2024; Park et al., 2024; Gani et al., 2022). One direct ViT study found that a DINO ViT-B/16 lost **over 70% ImageNet accuracy after only 10 iterations** of fine-tuning on CIFAR-100, and the authors argue the effect is magnified in ViTs relative to earlier networks (Bafghi et al., 2024). The same paper reports that PEFT variants such as Block Expansion and LoRA preserved more of the pretraining-domain performance than full fine-tuning while matching or beating downstream adaptation (Bafghi et al., 2024).

This instability is not unique to one benchmark. Few-shot and incremental settings repeatedly describe the same pair of failure modes: **catastrophic forgetting and overfitting** become the main barrier when only a few labeled examples are available (Park et al., 2024; Kim et al., 2022). In multimodal fine-tuning, moderate early updates can help, but continued fine-tuning on narrow data reduces performance on other datasets, and more diverse tuning data reduces forgetting (Zhai et al., 2023).

Why LP-FT Wins

LP-FT outperforms naive end-to-end fine-tuning because it first learns a **good classifier head**, so the later full-model updates distort pretrained features less (Kumar et al., 2022). Kumar et al. show that naive fine-tuning tends to move the feature extractor substantially while the randomly initialized head is still being fit, creating inconsistency between in-distribution and shifted examples (Kumar et al., 2022). NTK analysis reaches the same mechanistic conclusion: LP creates a near-optimal head, and the larger head norm at the start of fine-tuning reduces subsequent feature change (Tomihari & Sato, 2024).

Setting	FT	LP	LP-FT	Takeaway
STL	82.4	85.1	90.7	LP-FT best (Kumar et al., 2022)
CIFAR-10.1	92.3	82.7	93.5	Even small shift favors LP-FT (Kumar et al., 2022)
DomainNet	55.5	79.7	80.7	Large shift strongly favors LP-FT (Kumar et al., 2022)

Setting	FT	LP	LP-FT	Takeaway
FMoW	32.0	36.6	36.8	Robustness gain persists (Kumar et al., 2022)

FIGURE 1 LP-FT versus fine-tuning under distribution shift

LP-FT helps most when **pretrained features are already good** and the downstream shift is nontrivial (Kumar et al., 2022). When pretrained features are weak, full fine-tuning can recover more than linear probing alone, though LP-FT still tends to do best overall (Kumar et al., 2022). In extremely few-shot regimes, LP can beat full fine-tuning because updating the whole network overfits, but as sample size grows, full fine-tuning becomes more competitive (Kim et al., 2022). Related work in federated and 3D transfer settings also ties LP-FT's advantage to **reduced feature distortion**, especially under covariate or concept shift and partial feature overlap (Chen et al., 2025; Zhang et al., 2024).

PEFT Alternatives

The main alternatives are **LoRA**, **BitFit**, **adapters**, **VPT**, and **factorized variants** such as FacT and GLoRA (Ly & Nguyen, 2024; Jie & Deng, 2022; Chavan et al., 2023). LoRA is attractive because it tunes low-rank updates, transfers across modalities, and can avoid extra inference latency after reparameterization (Liu et al., 2024; Chavan et al., 2023). BitFit is simpler still, updating only bias terms, while adapters add bottleneck modules inside the ViT blocks (Ly & Nguyen, 2024; Marouf et al., 2023).

- **LoRA** can be strong and efficient, but it is not uniformly reliable; one ViT forgetting study found it underperformed on simpler datasets like CIFAR-10 (Bafghi et al., 2024).
- **Adapters** are flexible, but small adapters lose accuracy and their layer-wise sizing is hard to tune under storage constraints (Marouf et al., 2023).
- **BitFit and similar minimal updates** are cheap, but broader PEFT work suggests many such methods miss subtle high-frequency structure unless augmented, which matters in fine-grained tasks (Ly & Nguyen, 2024).

The most principled reason not to use these methods as the default here is that they mainly solve **parameter and memory efficiency**, not the core objective of preserving pretrained semantics while getting the benefits of full adaptation. Several PEFT methods can outperform full fine-tuning, sometimes with only **0.5%** or even **0.01%** trainable parameters, but their success is method- and hyperparameter-sensitive, and some introduce architecture choices or tuning burdens that are unnecessary if LP-FT already solves the main robustness problem (Jie & Deng, 2022; Hamdi et al., 2026; Marouf et al., 2023).

Across these three questions, the literature supports a simple hierarchy: naive full fine-tuning is most brittle on **small datasets**, LP-FT is the strongest **simple default** when pretrained features are good, and PEFT alternatives are useful when memory, storage, or multi-task deployment constraints matter more than choosing the least complicated robust fine-tuning recipe.

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